



Test Automation Framework (TAF::Perl)

Database Software Installation Quick Start

MariaDB Foundation <https://mariadb.org>

Open, vendor-neutral, community-driven tooling for database testing and benchmarking.

TAF-Perl is maintained and published by the MariaDB Foundation to support transparent, reproducible, and fair performance evaluation across the open source database ecosystem. The framework is part of the Foundation's mission to provide open tooling that benefits the entire database community.

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Introduction

The Test Automation Framework (TAF::Perl) is a vendor-neutral, open-source framework designed to support reproducible database software testing and benchmarking. It is maintained by the MariaDB Foundation to promote transparent evaluation across the open source database ecosystem.

This guide focuses on the installation lifecycle of database software within TAF. Install operations are central to the framework's ability to validate performance, compatibility, and initialization behavior across multiple database vendors and versions.

TAF supports a variety of install types, including:

- **Tarball installs** (classic unpacked binaries)
- **Systemd-targeted tarballs** (with layout normalization)
- **RPM bundles** (multi-package installs)
- **TAF install actions** (automated lifecycle operations)

TAF is designed to operate without polluting the host system. It avoids system service dependencies, root privileges, and global configuration changes. All installs are local, relocatable, and managed within the framework's directory structure. This ensures deterministic behavior, contributor-proof reproducibility, and safe parallel testing across multiple database versions.

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Database Software Installation

TAF provides two primary methods for installing database software into the framework. Both methods produce the same normalized installation layout and update the active install marker so that subsequent actions (setup-db, start-db, run-tests, build-client) operate consistently.

The two installation paths are:

1. Direct command-line installation
2. Action-driven installation (using a properties file and an action sequence)

Each method is described in the sections below.

1. Direct Command-Line Installation

Direct installation requires no properties file and no TAF action. This method is ideal for quick testing, verifying package compatibility, or preparing an installation before running a full action sequence.

Required Options

```
--db-software-install  
--db-software-install-packages=/path/pkg1.tar.gz[,pkg2.rpm,...]
```

Example: Installing a MariaDB Systemd Tarball

```
perl taf.pl --db-software-install \  
    --db-software-install-packages=/home/dl/mariadb/mariadb-12.1.2-linux-systemd-x86_64.tar.gz
```

Key output:

```
Install has completed. Happy benchmarking!  
Active install set to: ../mariadb-12.1.2-linux-systemd-x86_64
```

Verify the installation:

```
perl taf.pl --list-database-software-installs
```

Example: Installing a MySQL RPM Bundle

```
perl taf.pl --db-software-install \  
    --db-software-install-packages=/home/dl/mysql/8.0/*
```

TAF automatically detects the server-capable package:

```
_SelectBasePackage -> server-capable package: mysql-community-server-8.0.45...  
Active install set to: ../mysql-community-server-8.0.45-1.el8.x86_64
```

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Example: Installing a MariaDB RPM Bundle

```
perl taf.pl --db-software-install \  
    --db-software-install-packages=/home/dl/mariadb/mariadb-10.11.15-rhel-8-x86_64-rpms.tar
```

TAF identifies the bundle and normalizes the layout:

```
bundle detected: mariadb-10.11.15-rhel-8-x86_64-rpms.tar  
Active install set to: .../mariadb-10.11.15-rhel-8-x86_64-rpms
```

2. Action-Driven Installation

Action-driven installation uses a properties file and executes a full lifecycle sequence. This method is recommended for benchmarking, automated testing, and repeatable workflows.

Available Actions

- install-init-db-exit
- install-init-start-db-exit
- install-init-start-db-run-tests
- install-init-start-db-build-client-run-tests

Each action builds on the previous one, adding additional lifecycle stages.

```
perl taf.pl \  
    --properties=./properties/mysql/beta/sysbench_lua.properties \  
    --action=install-init-db-exit \  
    --db-software-install-packages=/home/dl/mysql/mysql-9.6.0-1.el8.x86_64.rpm-bundle.tar
```

Key lifecycle events:

- Package bundle detected
- Installation validated and normalized
- Database initialized
- Bootstrap server started for user provisioning
- Root and tester users created
- Bootstrap server shut down cleanly

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2.2 install-init-start-db-exit

Installs, initializes, starts the database, and exits.

```
perl taf.pl \  
--properties=./properties/mysql/beta/sysbench_lua.properties \  
--action=install-init-start-db-exit \  
--db-software-install-packages=/home/dl/mysql/mysql-8.4.8-1.el8.x86_64.rpm-bundle.tar
```

Key lifecycle events:

- Installation and initialization completed
- Runtime server started successfully
- Database left running (skip_database_shutdown=TRUE)

2.3 install-init-start-db-build-client-run-tests

Executes the full lifecycle: install, init, start, build client tools, and run tests.

```
perl taf.pl \  
--properties=./properties/mariadb/beta/sysbench_lua.properties \  
--action=install-init-start-db-build-client-run-tests \  
--db-software-install-packages=/home/dl/mariadb/mariadb-11.8.5-linux-systemd-x86_64.tar.gz
```

Key lifecycle events:

- Installation and initialization completed
- MariaDB runtime server started
- sysbench-lua client source built via CMake
- Test suite executed (sysbench-lua)
- Database shut down cleanly
- Logs archived under the run directory

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Active Install Marker

TAF maintains a small text file that records which database software installation is currently active. This ensures that all actions operate against the same installation.

The active install marker is updated when:

- A new installation is performed
- The user selects an install via `--set-active-database-software-install`
- The user specifies an install directory via `--db-software-install-dir`

To view the active installation:

```
perl taf.pl --list-active-db-install
```

```
== AVAILABLE DATABASE SOFTWARE INSTALLS ==
```

```
1: /home/taf-perl/database_software_installs/mariadb-10.11.15-rhel-8-x86_64-rpms
2: /home/taf-perl/database_software_installs/mariadb-11.8.5-linux-systemd-x86_64
3: /home/taf-perl/database_software_installs/mariadb-12.1.2-linux-systemd-x86_64
[ACTIVE] 4: /home/taf-perl/database_software_installs/mysql-8.4.8-1.el8.x86_64.rpm-bundle
5: /home/taf-perl/database_software_installs/mysql-9.6.0-1.el8.x86_64.rpm-bundle
6: /home/taf-perl/database_software_installs/mysql-community-server-8.0.45-1.el8.x86_64
```

To change the active installation:

From command-line without user's properties file being required.

```
O = --set-active-database-software-install
    : Mark a database software install as active.
```

From command-line or user's properties file where user's properties file is required.

```
O = --db-software-install-dir=<path>
P = taf.db_software_install_dir=<path>
    : Directory of an existing database software install.
    : Using this option also updates the active install marker,
      making the specified directory the active installation
      for all subsequent TAF actions (setup-client, setup-db,
      start-db, run-tests, etc.).
```

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Example of --set-active-database-software-install

```
perl taf.pl --set-active-database-software-install
```

Setting active database software install

Available installs:

```
1: /home/taf-perl/database_software_installs/mariadb-10.11.15-rhel-8-x86_64-rpms
2: /home/taf-perl/database_software_installs/mariadb-11.8.5-linux-systemd-x86_64
3: /home/taf-perl/database_software_installs/mariadb-12.1.2-linux-systemd-x86_64
[ACTIVE] 4: /home/taf-perl/database_software_installs/mysql-8.4.8-1.el8.x86_64.rpm-bundle
5: /home/taf-perl/database_software_installs/mysql-9.6.0-1.el8.x86_64.rpm-bundle
6: /home/taf-perl/database_software_installs/mysql-community-server-8.0.45-1.el8.x86_64
```

Enter selection (1..6, or 0 to exit): 3

Active install marker updated: /home/taf-perl/.taf-active-install.txt -> /home/taf-perl/database_software_installs/mariadb-12.1.2-linux-systemd-x86_64

Available installs:

```
1: /home/taf-perl/database_software_installs/mariadb-10.11.15-rhel-8-x86_64-rpms
2: /home/taf-perl/database_software_installs/mariadb-11.8.5-linux-systemd-x86_64
[ACTIVE] 3: /home/taf-perl/database_software_installs/mariadb-12.1.2-linux-systemd-x86_64
4: /home/taf-perl/database_software_installs/mysql-8.4.8-1.el8.x86_64.rpm-bundle
5: /home/taf-perl/database_software_installs/mysql-9.6.0-1.el8.x86_64.rpm-bundle
6: /home/taf-perl/database_software_installs/mysql-community-server-8.0.45-1.el8.x86_64
```

3. Differences Between Install Types

Database vendors ship their software in a variety of formats, each with its own directory layout, initialization behavior, and packaging conventions. TAF abstracts these differences by normalizing every installation into a consistent internal structure. This section highlights the key differences between install types and shows how TAF handles each one.

3.1 MariaDB Systemd Tarball Installs

MariaDB provides systemd-targeted tarballs that include systemd unit files and a layout intended for system-level deployment. TAF does not use systemd and therefore ignores all systemd-specific components.

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Characteristics

- Includes systemd unit files and systemd-oriented directory layout
- Binaries may be located under `bin/` or `sbin/` depending on build
- Includes systemd service definitions that TAF does not use
- Initialization uses the vendor-provided `--initialize path`

How TAF Handles It

- Ignores systemd units entirely
- Normalizes the installation into `database_software_installs/`
- Ensures all binaries are reachable under the normalized layout
- Runs initialization using the correct vendor-specific binary
- Updates the active install marker automatically

Example (from testing)

```
perl taf.pl --db-software-install \  
--db-software-install-packages=mariadb-12.1.2-linux-systemd-x86_64.tar.gz
```

Install has completed. Happy benchmarking!

Active install set to: `.../mariadb-12.1.2-linux-systemd-x86_64`

3.2 MariaDB Non-Systemd Tarball Installs

Non-systemd tarballs follow the classic MariaDB layout. These are simpler and do not include systemd units.

Characteristics

- Classic tarball layout
- Binaries typically under `bin/`
- No systemd service files
- Initialization behavior may differ from systemd builds

How TAF Handles It

- Detects the tarball type automatically
- Normalizes the layout into the TAF directory structure
- Ensures all binaries are reachable
- Runs initialization using the appropriate vendor binary
- Updates the active install marker

Example (from testing)

bundle detected: `mariadb-10.11.15-rhel-8-x86_64-rpms.tar`

Active install set to: `.../mariadb-10.11.15-rhel-8-x86_64-rpms`

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3.3 RPM Bundle Installs (MariaDB and MySQL)

RPM bundles contain multiple RPM packages (server, client, libs, plugins). These are already structured and do not require system-level installation when used with TAF.

Characteristics

- Multi-package bundles
- Contain server-capable RPMs and supporting RPMs
- Designed for system installation, but TAF extracts them locally
- No system pollution (no root, no systemd, no global paths)

How TAF Handles It

- Detects the server-capable RPM automatically
- Extracts all RPMs into a temporary staging area
- Normalizes the layout into the TAF install directory
- Ensures all binaries are reachable
- Updates the active install marker

Example (from testing)

`_SelectBasePackage -> server-capable package: mysql-community-server-8.0.45...`
Active install set to: `.../mysql-community-server-8.0.45-1.el8.x86_64`

3.4 TAF Install Actions

TAF actions provide a fully automated lifecycle: install, initialize, start, build clients, and run tests. This allows users to perform complete database setup without touching the command line beyond the initial invocation.

Characteristics

- Requires a properties file
- Executes a defined lifecycle sequence
- Performs version detection and capability detection
- Normalizes layout automatically
- Handles initialization, startup, client builds, and test execution

How TAF Handles It

- Detects package type (tarball, systemd tarball, RPM bundle)
- Validates version and capabilities
- Normalizes the installation
- Loads the correct database plugin
- Initializes and starts the database
- Builds client tools when required
- Runs the selected test suite
- Archives logs and results

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Example (from testing)

TAF ACTION: install-init-start-db-build-client-run-tests

...

Database started successfully.

SYSBENCH-LUA -> BuildClient -> Complete

TAF::Run::RunTests TestSuite = sysbench-lua

...

TAF Exit Code: 0

Conclusion

This Quick Start guide introduced the core concepts and workflows for installing database software within TAF::Perl. By walking through direct command-line installs, action-driven lifecycle installs, and the differences between package types, you now have a complete understanding of how TAF normalizes vendor-specific layouts into a consistent, reproducible structure.

TAF's installation subsystem is designed to be:

- vendor-neutral
- non-intrusive
- deterministic
- fully reproducible

Every installation — whether from a tarball, systemd tarball, or RPM bundle — is isolated within the framework and tracked through the active install marker. This ensures that all subsequent operations (initialization, startup, client builds, and test execution) run against the correct database software without requiring system-level configuration or root privileges.

With these fundamentals in place, you are ready to explore the broader capabilities of TAF, including:

- running full benchmark suites
- building and validating client tools
- comparing database versions and configurations
- automating repeatable performance evaluations
- archiving and analyzing test results

TAF is built to support transparent, fair, and community-driven testing across the open source database ecosystem. The workflows demonstrated in this guide form the foundation for more advanced usage, and all higher-level operations build on the installation principles covered here.

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End of Document

This Quick Start was prepared and published by the MariaDB Foundation <https://mariadb.org>

TAF::Perl is part of the Foundation's commitment to open, vendor-neutral tooling for the database ecosystem. The framework is maintained to support fair benchmarking, reproducible testing, and transparent engineering practices across all database makers.

For contributions, feedback, or questions, please visit the Foundation website or contact the maintainers directly.

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